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Core Skills

Designing, building and operating an MLOps platform on on-prem Kubernetes:

I codified the entire path — ingestion → lakehouse → transformation → lineage → distributed training → experiment / model registry — on bare-metal Kubernetes in **Pulumi (Python IaC)**, and I build and operate it alone. Across networking (Cilium), storage (SeaweedFS) and identity (Dex OIDC → Kubernetes RBAC) I **authored the majority of all 12 stacks** (93% of stack-code changes) and run **10 nodes / 235 running pods / 23 internal services** in production.

Data pipelines and the lakehouse:

Designed and implemented a medallion pipeline on **Airflow 3.x + dbt (Cosmos) + Apache Iceberg** with lineage auto-recorded in **DataHub**; authored 83% of the ETL repository and 93% of its shared framework. Previously ran large-scale cloud pipelines on AWS EMR / PySpark.

Architecture and trade-off decisions made by measurement:

I measure before choosing tools. Turned a GPU purchase request into reclamation with 10 days of telemetry (**5.3% utilization, zero queued jobs, 1,112 GPU-hours held idle**), cut training-data delivery **25×** by redesigning network and storage, and designed a reproducibility contract from a quantified survey of the research workflow. Weighing cost, performance and operational complexity is the same habit that drove earlier decisions such as adopting PGVector.

AI agents run continuously, backed by verification gates:

I run AI coding agents continuously to lift **one person's throughput to team scale** (678 merged PRs in 6 months), and keep that speed honest with **719 SDK unit tests**, CI gates and pre-commit hooks that block tag/version drift and private-key commits, and **23 incident write-ups and 30 PRDs**. Earlier, built deduplication systems and chatbot prototypes with LangChain and embedding models.

Full-stack breadth and team leadership:

A path from iOS through backend (FastAPI / Flask / Spring Boot), frontend (Next.js) and cloud (AWS / GCP) let me lead a data engineering team; today I design and enforce the split where **researchers own domain ETL and the platform owns execution, lineage and verification**.

Technical Skills

Languages: Python | Swift | JavaScript | Objective-C | Shell | SQL

Frameworks: Flink CDC | PySpark | Flask | FastAPI | Airflow | Kafka | Langchain | OpenAI | Streamlit | ReactJS | NextJS | Bun

iOS: ReactorKit | Async Await | Alamofire | Snapkit | Crashlytics | Kingfisher | RxDataSource | SceneKit | SPM | React-Native

Database: ClickHouse | MariaDB | PostgreSQL | Firestore | Neo4J

DevOps: uv | Ubuntu | Docker | AWS (EC2, Lambda, S3, IAM, EMR, Route53, RDS Aurora), Firebase (Firestore, Storage), GCP (Google Cloud Run, Artifact Registry)

LLM: Langchain | LangGraph | OpenAI(Text Embedding) | RAG(Neo4J, PostgreSQL) | Prompt Engineering

ML: HuggingFace | DVC | NLP(Entity Recognition)

VectorStore: PGVector

Data Analytics: K-Means | PCA | CytoscapeJS | SigmaJS

Career Summary

Data Engineer (approx. 3 years): Deep experience in designing, building, and optimizing large-scale data pipelines in a cloud (AWS) environment. Led the development of data platforms and MLOps pipelines using technologies such as PySpark, Airflow, and Vector DBs.

Backend Engineer (cumulative approx. 2-3 years): Experienced in developing API servers and management systems using Python (Flask, FastAPI). Built various backend systems, including those for real-time communication with WebSockets and Firebase integration.

Frontend & Client (approx. 4 years): Experienced in developing data visualization dashboards and BI tools with Next.js. Started career as an iOS Developer, designing reactive architectures based on ReactorKit and building native applications.

Experiences

HUINNO

Dec 2025 - Present

Data Engineer

Seoul

- **Designed, built and operate the company's entire MLOps pipeline** (ingestion → lakehouse → transformation → lineage → distributed training → experiment / model registry). **Sole owner of cluster construction on a two-person platform team** — networking, storage, catalog, lineage, orchestration, training and identity: **authored the majority of all 12 Pulumi stacks** (93% of stack-code changes; 6 stacks above 96%). Operated **10 nodes / 235 running pods / 23 internal services** in production over 6 months, shipping **678 merged PRs and 1,204 commits (92% authored personally)**, and codified bare-metal **Kubernetes** end-to-end in **Pulumi (Python IaC)** for reproducible deploys.
- **Designed and implemented every branch of the ETL pipeline** (separate repository; **83% of 1,599 commits** authored personally). Wrote **93%** of the shared cross-product framework and **100%** each of the CAST, MIMIC-IV and operational-maintenance pipelines, and designed the remaining products to extend that framework — establishing a split where **researchers own domain ETL while the platform owns execution, lineage, verification and retries**.
- **[Data] Removed the single-person dependency in dataset preparation**. Previously a researcher had to ask one data scientist, who ran a hard-coded script over SSH for **hours** and handed back a file — work stalled whenever that person was away, and reproducibility, lineage and versioning were all absent. Moved Bronze / Silver / Gold transformation to **scheduled execution with Airflow 3.x + dbt (Cosmos) + Apache Iceberg** and auto-recorded table-level lineage in **DataHub**, so researchers query directly instead of requesting and waiting.
- **[Training] Built the distributed-training foundation to replace single-GPU manual runs**. Unified **10 GPUs** of mixed SKUs into one **KubeRay** pool behind a **single worker image** with pinned CUDA / PyTorch / ONNXRuntime and an internal SDK submission API (automatic checkpointing and mid-run resume supported). **Adoption is in progress; migrating research code remains open** — the two items below are the measurement work that identified what blocks it.
- **[Design] Quantified the existing research workflow and designed a reproducibility architecture from the findings**. Surveyed the research repository and measured **51 remote branches, 67-88% of notebook cells carrying stored outputs, and experiment records isolated in local folders**; identified the root cause as not missing tooling but **results having no coordinates**. Designed and implemented a reproducibility contract that stamps **version, content hash, code commit, environment lock and seed** onto a dataset at execution time (freeze tool, manifest schema, integrity-verification CLI), positioning MLflow tracking / model registry (S3 artifacts, SSO, DataHub mirror) as the store for those coordinates.
- **[Foundation] Cut training-data delivery time 25×** — redesigned network and storage from **93.7 Mbps to 2.5 Gbps** (iperf3 measured), taking an 8 TB transfer from **8.4 days to 7.8 hours**. Applied **NVMe / HDD tiering** to an Iceberg lakehouse on **SeaweedFS** (S3-compatible), and opened Iceberg to ad-hoc SQL via **PostgreSQL + pg_duckdb**, replacing hand-written pyiceberg code with a **single line of browser SQL**.
- **[Access] Consolidated shared passwords and unauthenticated endpoints into one identity**. Applied SSO across **23 internal services** with **Dex (GitHub OIDC) + oauth2-proxy** and carried the same identity through to **Kubernetes RBAC**. Replaced direct SSH into GPU servers with **browser-based JupyterHub development environments** (per-user volumes, egress-policy isolation, one-command bootstrap).
- **[Judgment] Reversed a GPU procurement decision with measurement**. Asked to evaluate expansion, measured 10 days with DCGM / Prometheus and found **5.3% utilization with zero queued jobs** — the bottleneck was not capacity but **1,112 GPU-hours held idle by dormant sessions (91% of all held time)** — and redirected the decision from purchase to reclamation.
- **[Method] Ran AI coding agents continuously to lift one person's throughput to team scale, while backing that speed with verification gates**. Delivered the above in 6 months alongside **719 SDK unit tests, CI gates and pre-commit hooks** that block tag/version drift and private-key commits, and **23 incident write-ups and 30 PRDs** that turn failures into prevention procedures.

Cellkey

Jul 2025 - Dec 2025

Data Engineer

Seocho-gu, Seoul

- Ran a **PoC** of a shared-nothing **StarRocks** cluster on AWS EC2, measuring an **80%** reduction in analytical query time (7s → 1.4s). Evaluation stage; not promoted to production.
- Designed and deployed a local data refinement pipeline using **Apache Airflow (3.x)** for automated data processing.
- Maintained and optimized **AWS resources** including Lambda, Batch, and Step Functions for scalable data workflows.

Naepopquiz

Mar 2025 - Jun 2025

Back-End Engineer

Freelance

- Implemented Android & iOS In-App Purchase API (subscriptions, renewals, cancellations)
- Deployed a Flask-based Dockerized web server on Google Cloud Run
- Implemented a feature to verify ownership of Firebase resources
- Wrote unit test cases for LLM-driven API routes, services, and functions
- Operated Firebase Firestore and Cloud Storage
- Refactored legacy code

Eazel Inc.

Jun 2022 - Sep 2024

Team Lead, AI Labs

Jung-gu, Seoul

- Oversaw the development and operation of the art market ETL data pipeline; as team lead, led architecture design, deployment, monitoring, and performance optimization
- Built dashboards based on Flask/NextJS
- Developed and operated auction data scrapers
- Implemented periodic data collection, purification, and loading pipelines using Airflow
- Implemented data purification and loading pipelines using PySpark applications on AWS EMR
- Developed a data deduplication system using OpenAI/Facebook embedding models and improved its accuracy/speed (e.g., by applying a vector DB)
- Managed AWS services (EC2, IAM, S3, Route53, RDS, EMR)
- Implemented CI/CD (Continuous Deployment) using Github Actions
- Implemented a Chatbot Prototype using LangChain and Streamlit
- Led the implementation of a Neo4J Cypher query tool and a PostgreSQL query tool for a chatbot
- Implemented data scraping using LangChain and Information Extraction Prompts

Eazel Inc.

Oct 2021 - Jun 2022

MLOps Engineer, MLOps

Jung-gu, Seoul

- Experienced in the end-to-end process from data analysis to modeling and validation (utilizing HuggingFace NLP, etc.)
- Developed and operated Instagram data scrapers
- Developed a social network analysis dashboard based on Flask/NextJS/CytoscapeJS and derived insights (contributed to securing Series A funding)
- Implemented the ML Lifecycle using DVC
- Managed data using PostgreSQL

Eazel Inc.

Jan 2019 - Oct 2021

iOS Developer, iOS Team

Gangnam, Seoul

- Led the design of the API client and integrated backend logic, considering iOS app performance and scalability
- Introduced CI/CD with Github Actions
- Led the introduction of ReactorKit-based reactive programming and its architecture design, improving app responsiveness and maintainability

Finple Inc.

Jan 2018 - Nov 2018

Software Engineer, DevOps

Yeongdeungpo, Seoul

- Maintained hybrid iOS and Android apps
- Developed a NodeJS-based data scraper

Projects

Eazel Inc. Data Engineer, AI Labs	Jun 2022 - Sep 2024 Jung-gu, Seoul
Development and Operation of Data Pipeline for Price Database Service • Built data pipeline using a single AWS Aurora RDS to save development resources. • Processed and loaded data using PySpark Applications on AWS EMR. • Implemented partitioning and data optimization for distributed processing. • Key contribution to Series B funding by leading the overall design and construction of the data platform (Price Database/MLOps).	Jan 2023 - Oct 2023
Development and Operation of Art Market Data Scraper • Developed data scraper using requests and BeautifulSoup4. • Developed high-speed scraper utilizing MultiThreading and MultiProcessing. • Developed structured data scraping technique using Langchain's Information Extraction Prompt. • Developed a social relationship collection scraper written in Swift, operating in the iOS environment. • Managed multiple scraper instances (50 simultaneously) using boto3. • Managed scraper instances by connecting to the scraper management server using WebSockets.	Jan 2023 - Feb 2023
Development and Operation of Central Management Server for Data Scrapers • Developed API endpoints using Flask for central management of scrapers. • Connected with scrapers via WebSocket to pass target information for scraping. • Transmitted Login Credentials using SHA-256 encryption. • Developed API Endpoints using Flask Blueprints. • Implemented reverse proxy using Nginx. • Managed HTTPS certificates using Route53 and Certbot.	Jan 2023 - Feb 2023
Development and Operation of BI Tool • Developed server for BI tool using Flask and NextJS. • Developed API to deliver social relationship data. • Developed BI Page using Cytoscape and SigmaJS to display social relationships in a directed graph format. • Developed API for data labeling. • Developed Components using NextJS and Carbon Design System. • Managed HTTPS certificates using Route53 and Certbot. • Deployed to EC2 Instance (Ubuntu 22.04) using Docker Compose.	Nov 2023 - Nov 2023
Deployment and Operation of Airflow Service • Created scraping scheduling DAGs using Airflow's PythonOperator to communicate with the central scraper management server. • Created data refinement and loading scheduling DAGs using AWS EMR service via boto3. • Operated DAGs using Sequential Executor. • Managed HTTPS certificates using Route53 and Certbot. • Deployed to EC2 Instance (Ubuntu 22.04) using Docker Compose.	Jan 2023 - Feb 2024
Development of Data Deduplication Feature • Collaborated with ML Research Engineer to develop a deduplication system by calculating distances between Embedding Vectors using Faiss-CPU with IVFFlat Index. • Extracted Image Embedding Vectors using Facebook's SSCD model. • Extracted Text Embedding Vectors using OpenAI's Text Embedding model Ada. • Reduced index creation overhead by 50% using Postgres's PGVector 0.5.0 Extension. • Created PGVector's IVFFlat and HNSW Indexes. • Migrated data from existing Numpy.array to pgvector::vector.	Nov 2023 - Feb 2024
Data Warehouse Schema Design for Efficient Data Management • Designed a data warehouse with a Single Source of Truth table using DBDiagram to efficiently manage duplicate data collected from various sources.	Feb 2024 - Sep 2024

- Designed a data warehouse schema using S3 as a data lake to reduce costs incurred from using existing RDS as both data source and data warehouse storage.
- Designed DataSource, DataWarehouse, DataMart, and SSoT schemas, encompassing approximately 50 tables.

AWS Service Management

- Implemented user-specific service access control based on IAM Roles.
- Migrated EC2 x86 instance types to arm64 for cost savings.
- Operated Aurora RDS PostgreSQL with separate Read and Write Instances.
- Selected services by weighing resource costs between Managed services and EC2 services.

Eazel Inc.

MLOps Engineer, MLOps

Oct 2021 - Jun 2022

Jung-gu, Seoul

Art Market Analysis PoC

Apr 2022 - Jun 2022

- Quantitatively analyzed and refined artist CV data using Entity Recognition technology with HuggingFace's NLP.
- Analyzed articles using Word Histograms.
- Analyzed articles using Sentiment Analysis.

Instagram Relationship Data Analysis

Feb 2022 - Apr 2022

- Analyzed relationships using Node2Vec, K-means, and PCA techniques.
- Developed data visualization page showing directed graph relationships using NextJS, Flask, and CytoscapeJS.
- Secured Series A funding based on analysis data.

Instagram Relationship Information Scraper Development

Oct 2021 - Jun 2023

- Devised and implemented various methods for continuous data collection.
- Developed dynamic page scraper using Selenium.
- Developed data scraper via Public API using requests.
- Developed data scraper using Instagrapi, an Open Source Project utilizing Instagram's private API.
- Migrated Instagrapi (written in Python) to Swift language to develop a data scraper operating in a mobile environment
- Contributed to Open Source project for social relationship data collection: [Instagrapi](#)

Art Artist Information Scraper Development

Oct 2021 - Nov 2021

- Developed static page scraper using requests.
- Developed high-speed scraping environment by running multiple EC2 instances using boto3 and MultiThreading.
- Collected information on 600,000 artists and 3 million artist CV entries.

PostgreSQL-based Collected Data Management

- Stored Raw Data and Purified data separately through schema separation.
- Refined data using Python and psycopg2

Eazel Inc.

iOS Developer, iOS Team

Jan 2019 - Oct 2021

Gangnam, Seoul

iOS Application Development and Deployment

- Built CI/CD using Fastlane and Github Actions.
- Implemented Artwork Hanger feature using ARKit to hang artwork on walls.
- Implemented Push Notification feature.
- Built API Layer using Alamofire to consume Backend APIs.
- Integrated Unity's Google Cardboard VR feature using Unity Bridge.
- Implemented feature to download assets stored in AWS S3 for VR functionality.
- Implemented Download Service to support multiple downloads.

- Used ReactorKit to implement Reactive Pattern Programming, implementing Model, View, ReactorKit architecture.
- Established unit test environment and progressively expanded coverage.
- Implemented Autolayout using SnapKit.
- Developed UIImageView using Kingfisher.
- Developed Pinterest-style UICollectionView.

Finple Inc.	Jan 2018 - Dec 2018
Software Engineer, DevOps	Yeongdeungpo, Seoul
iOS Hybrid Application Maintenance	Jan 2018 - Nov 2018
• Maintained iOS Application using UIWebView and WKWebView with Alert-based communication employing VueJS-based Web views. • Migrated from Objective-C to Swift. • Developed features corresponding to ActionCodes transmitted via Alerts.	
Android Hybrid Application Maintenance	Jan 2018 - Nov 2018
• Maintained Android Application using ChromeWebView with Alert-based communication employing VueJS-based Web views. • Maintained using Java. • Developed features corresponding to ActionCodes transmitted via Alerts.	
NodeJS-based Data Scraper Development	Jan 2018 - Nov 2018
• Implemented login functionality using pre-existing encryption/decryption module. • Implemented functionality to scrape specific pages with pagination.	

Education

Coursera	Feb 2023
<u>Convolutional Neural Networks</u>	
Coursera	Nov 2022
<u>Structuring Machine Learning Projects</u>	
Coursera	Oct 2022
<u>Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization</u>	
Coursera	Sep 2022
<u>Neural Networks and Deep Learning</u>	
Engineer Information Processing	Nov 2017
Human Resources Development Service of Korea	
Languages	
• English: AL (OPIC) • Korean: Native	
Korea Polytechnic University	Mar 2010 - Feb 2018
B.S., Computer Engineering (GPA: 3.4 / 4.5) (Major GPA: 3.73 / 4.5)	Siheung, Gyeonggi-do
• Relevant Coursework: Data Structures, Algorithms (C), Operating System, Computer Architecture, Clouding Computing • <u>[Probono] Furniture Arrangement System Using Augmented Reality for the Socially Vulnerable</u>	

Other Activities

Guryong Village Coal Briquette Volunteering	Jan 2018
Participated in Let's Us Go iOS Hackathon	Jun 2024
Developed a layer using OpenAI API to receive standardized responses for food recipes	
Republic of Korea Army, Sergeant (Professional NCO)	Jun 2011 - Sep 2013
Honorable Discharge. Created an XLSX payroll calculator using Visual Basic Macro	

In-House MLOps Platform on Bare-Metal Kubernetes (Pulumi IaC)

- **Core Problem & Goal:** Build an in-house **MLOps platform** from scratch on bare-metal resources so ML researchers can submit distributed training jobs and data engineers can run lakehouse / analytics workloads reliably. The goal was to operate **storage, catalog, lineage, orchestration, distributed training, and observability consistently in a single cluster**, without depending on external managed services.
- **My Role:** Sole owner across the full stack **designing, building, and operating** every layer on a multi-node kubeadm cluster (GPU / storage / CPU node pools) - IaC, networking, HA, lakehouse, orchestration, distributed training, observability.
- **Problem-Solving Process:**
 - * **Trade-off (Managed vs. Self-hosted):** Managed K8s / managed lakehouse offers a fast start, but given internal resource and security requirements plus long-term cost, **bare-metal self-hosted** was the superior choice. Accepted higher up-front investment to keep every layer in-cluster.
 - * **Implementation (How):** Modularized infrastructure into **12 Pulumi stacks** (platform → base → storage → catalog → lineage → orchestration → staging → compute → mlops → cicd → identity → devenv) with explicit cross-stack references for reproducible single-command deploys. Built networking on **Cilium CNI** (KPR + Gateway API + LB-IPAM + L2 Announcements); exposed *.<internal>.* hostnames via a single Gateway with in-cluster CA HTTPS.
- **Results & Impact:**
 - * **Reproducible IaC:** Infrastructure changes are tracked via PRs; new environments (e.g., separate staging) can be reconstituted instantly from stack combinations.
 - * **Unified operations:** All hostnames exposed through one Gateway centralizes certificate and routing management, and minimizes external dependencies.

Active-Active HA Architecture for SPoF Components (CoreDNS, Airflow scheduler)

- **Core Problem & Goal:** Training data pipelines must **keep running through single-node failures**. Critical in-cluster components (CoreDNS, Airflow scheduler) running as single instances meant any node fault could halt the entire workload.
- **My Role:** Identified SPoF candidates and designed / introduced **multi-node active-active redundancy policies** per component; codified operations into runbooks.
- **Problem-Solving Process:**
 - * **Why active-active:** Active-passive failover has unavoidable detection / failover lag, and our training pipelines have **expensive catch-up costs after a restart**. Chose multi-instance **active-active** even at the cost of more operational complexity.
 - * **Implementation (How):** Ran SPoF components (CoreDNS, Airflow scheduler) as multi-replica deployments; applied **PriorityClass / PodDisruptionBudget (PDB)** policies cluster-wide so at least one replica stays alive across node drains / upgrades. Documented operational procedures (drain order, scheduler-restart checks, lakehouse integrity checks) into **runbooks** so the next on-call engineer can follow the same path immediately.
- **Results & Impact:**
 - * **Uninterrupted operations through node failure:** Training DAGs and dataset sync continue through single-node faults and routine upgrades.
 - * **Operational knowledge captured:** Runbooks shorten the learning curve for hand-offs and let recurring incidents be resolved with minimal lead time.

SeaweedFS + Apache Iceberg Lakehouse (Bronze / Silver / Gold Medallion)

- **Core Problem & Goal:** Unify raw ingestion, refinement, and serving of training / analytics data in a single lakehouse. The goal was to assemble an **on-prem S3-compatible distributed storage + table format** lakehouse without external cloud dependency.
- **My Role:** Full ownership of the lakehouse layer - storage selection & provisioning, Iceberg medallion architecture, and ad-hoc query enablement.
- **Problem-Solving Process:**
 - * **Trade-off (MinIO vs. SeaweedFS):** Both are S3-compatible, but the need to manage hot / cold data efficiently across **NVMe / HDD disk tiers** and multi-node replica volume groups was a more natural fit for **SeaweedFS**.

- * **Implementation (How):** Built an **Apache Iceberg lakehouse** on SeaweedFS with a **Bronze / Silver / Gold medallion architecture** in production. To satisfy ad-hoc analytics needs, enabled direct SQL access to Iceberg via **PostgreSQL + pg_duckdb**, and deployed an analyst-facing **DuckDB UI as an in-cluster Pod** (nginx basic auth + automatic token injection) - collapsing the previous "write pyiceberg code" step into a one-line browser SQL query.
- **Results & Impact:**
 - * **Faster analytics cycles:** Analysts no longer need to set up pyiceberg / Spark environments - they run ad-hoc SQL directly in the browser.
 - * **Storage cost efficiency:** Hot data lands on NVMe, cold archival data on HDD tiers, balancing cost and performance in a single cluster.

Airflow 3.x + dbt + Cosmos Orchestration / Transformation Pipeline

- **Core Problem & Goal:** Operationalize Silver / Gold transformations plus Iceberg compaction and integrity verification as **standardized workflows**. Run dbt models naturally inside Airflow and separate prod / staging environments so **minor / patch upgrades ship without release risk**.
- **My Role:** Owned the entire path - Airflow Helm deployment, custom image, dbt + Cosmos integration, environment split, and upgrade standardization.
- **Problem-Solving Process:**
 - * **Why KubernetesExecutor:** Per-task pod isolation simplifies resource management and avoids dependency conflicts. Bundled the in-house dbt project + internal SDK into the same image so **every task runs against an identical dependency graph**.
 - * **Implementation (How):** Deployed Airflow 3.x via **Helm + KubernetesExecutor** on split prod / staging stacks, with **git-sync CD** and a custom image bundling dbt + Astronomer Cosmos + internal SDK. Wrapped Silver / Gold dbt models into Airflow TaskGroups; automated Iceberg compaction and integrity-verification DAGs. Standardized all changes to the **staging-first → prod-sequential** pattern.
- **Results & Impact:**
 - * **Automated transformation pipeline:** Integrity / compaction tasks moved from manual operations into DAGs, removing room for human error.
 - * **Safe upgrade path:** Minor upgrades of Airflow, dbt, or dependencies can ship without downtime via staging-first validation.

Distributed Training on Heterogeneous GPUs with KubeRay

- **Core Problem & Goal:** The cluster mixes **heterogeneous GPU node SKUs**; ML researchers should be able to **submit distributed training jobs without re-tooling their environment** every time.
- **My Role:** Ray cluster setup, unified training image, GPU-type-aware worker group policy, and automated in-house SDK installation pipeline.
- **Problem-Solving Process:**
 - * **Why a unified image:** Letting users set up environments per node causes recurring CUDA / dependency conflicts. **Unifying the worker image** and splitting only by GPU type via worker groups minimized operational cost and user friction.
 - * **Implementation (How):** Configured **KubeRay** over heterogeneous GPU nodes with a single worker image (CUDA 12.8 · Python 3.12 · PyTorch · ONNXRuntime · Triton); split worker groups by GPU type and automated internal training SDK installation so researchers submit jobs without code changes.
- **Results & Impact:**
 - * **Environment setup cost eliminated:** The setup step before distributed training disappeared; researchers focus on the code.
 - * **Asymmetric resource utilization:** Different GPU SKUs are jointly utilized inside a single Ray cluster, improving overall cluster utilization.

Integrated Catalog & Observability with DataHub / MLflow / Prometheus

- **Core Problem & Goal:** Dataset catalog, ML experiments / model registry, and cluster metrics were scattered. Needed a **unified observability layer where data flow and model lineage can be traced in one place**.
 - **My Role:** DataHub deployment + OpenLineage integration, MLflow operation, Prometheus / Grafana cluster-metric standardization, self-hosted Docker Registry + GitHub Actions CI/CD.
 - **Implementation (How):** **DataHub** for dataset catalog + **OpenLineage**-driven Airflow / dbt lineage collection; **MLflow** for experiments and model registry; **Prometheus + Grafana** for cluster metrics; **self-hosted Docker Registry + GitHub Actions** for unified image build and deploy.
 - **Results & Impact:**
 - * **Data + model lineage visibility:** Flow from dataset to model is traceable in a single view; change-impact analysis moves from manual tracing to catalog search.
 - * **Simplified image supply chain:** Infra and application image build / deploy unified into a single CI workflow with the self-hosted Registry.
-

Integrated Data Platform for Art Investment Analysis

- **Core Problem & Goal:** All art market data, including **artists, artworks, auctions, social networks, institutions, and artist CVs**, was fragmented across multiple sources. The final goal was to **build a reliable, integrated data pipeline** to encompass all this data, utilize it as an AI and statistics-based analysis platform, and **contribute to securing Series B funding**.
- **My Role:** As the Data Engineering Team Lead, I led the entire project, from the data pipeline's **architecture design to its implementation, operation, and the development of an LLM agent prototype**.
- **Problem-Solving Process:**
 - * **Trade-off Analysis (Pandas vs. PySpark):** Since the project's final goal was to collect all data, I **designed the architecture assuming** the high probability that the data scale would grow to hundreds of millions of records in the future. To secure future **scalability and reliability** over current development speed, I made the decision to proactively **adopt PySpark**, despite the higher initial overhead.
 - * **Specific Implementation (How):** Implemented the data purification and loading pipeline by having the **Python files that define each data processing stage included and executed sequentially as EMR Steps** when the AWS EMR cluster is instantiated. Within this pipeline, I built a system that utilized Facebook's **SSCD** and OpenAI's **Embedding** models and filtered duplicate data using **PGVector's HNSW** index.
- **Result & Impact:**
 - * **Business Impact:** The high-quality, integrated data platform that was built became a core company asset and played a **decisive role in successfully securing Series B funding**.
 - * **Technical Achievement:** By proactively adopting PySpark, I established the foundation for a scalable architecture that could stably operate the pipeline without a separate migration, even as data volume surged.

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Central Management & Monitoring Server for Large-Scale Data Scrapers

- **Core Problem & Goal:** As we operated **over 50 scraper instances simultaneously** for data collection, managing the status of each instance individually consumed significant resources. The goal was to **maximize the stability and operational efficiency of the data collection pipeline** by building a server to centrally control and monitor all scrapers in real-time.
- **My Role:** I was responsible for the **entire End-to-End development**, from the server's architecture design to API endpoint development, WebSocket communication, and Nginx/HTTPS deployment.
- **Problem-Solving Process:**
 - * **Technical Consideration (Why):** Judged that a **WebSocket** approach, which maintains a persistent bidirectional channel, was more efficient for real-time monitoring than HTTP Polling, as it reduces server load and propagates status changes instantly.
 - * **Trade-off Analysis (Trade-off):** Considered 'Flask' and 'FastAPI'. As the project's core need was stable real-time communication over complex async logic, I **chose 'Flask'**, which I was more familiar with and had a more mature library ecosystem (Flask-SocketIO), to prioritize development speed and stability.
 - * **Specific Implementation (How):** Developed scraper control APIs based on '**Flask**' and modularized the code with '**Flask-Blueprint**'. Implemented real-time status/metric exchange via '**WebSocket**'. Secured login credentials via '**SHA-256**' encryption and applied an HTTPS certificate from '**Certbot**' using '**Nginx**' as a reverse proxy.
- **Result & Impact:**
 - * **Maximized Operational Efficiency:** Dramatically reduced operational resources by enabling central management of over 50 scraper instances via a single API, which were previously managed manually.
 - * **Improved Stability & Reliability:** Significantly improved the data collection pipeline's overall stability and reliability by enabling immediate identification and remote restart of failed instances.

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Full-Stack BI Tool for Social Network Analysis and Data Purification

- **Core Problem & Goal:** The complex graph data from ML models was difficult for non-developer art experts to analyze. An efficient tool was needed for them to visually explore the network and simultaneously label data to improve its quality.
- **My Role:** Led the **entire Full-Stack development**, from data structure design and API development (**Flask**) to frontend (**Next.js**), visualization (**Cytoscape/SigmaJS**), and final deployment (**Docker, Nginx**).
- **Problem-Solving Process:**

- * **Technical Consideration (Why):** The key challenge was smoothly rendering large-scale graphs in a browser. I initially used **Cytoscape.js**, but migrated to the WebGL-based **Sigma.js** to improve performance when the number of nodes and edges grew.
 - * **Trade-off Analysis (Trade-off):** Chose IBM's **Carbon Design System** over more common UI libraries. Despite an initial learning curve due to fewer resources, I judged it would improve long-term productivity with its specialized components for data visualization dashboards.
 - * **Specific Implementation (How):** Developed RESTful APIs for social network and labeling data with **Flask**. Built the frontend with **Next.js** and UI components with **Carbon Design System**. Finally, containerized and deployed the Nginx, Next.js, and Flask servers on an EC2 instance using **Docker Compose**.
 - **Result & Impact:**
 - * **Improved Work Efficiency:** Dramatically reduced the time required for analysis by enabling art experts to directly explore social network data through the BI tool, instead of requesting it from developers.
 - * **Improved Data Quality:** Provided a feature for experts to directly correct and label data, which contributed to improving the quality of training data for the ML model and created a foundation for a virtuous cycle of data purification.
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Cost-Effective Airflow Implementation for Data Pipeline Automation

- **Core Problem & Goal:** The manual operation of the data collection and purification pipeline was inefficient and prone to errors. The goal was to build a stable scheduling system to **automate execution, monitoring, and reprocessing** while **minimizing operational costs**.
 - **My Role:** I was responsible for the **entire workflow automation process**, including Airflow infrastructure design, EC2/Docker-based deployment, and authoring core DAGs.
 - **Problem-Solving Process:**
 - * **Trade-off 1 (Infrastructure):** Chose to **deploy Airflow directly on EC2 with Docker Compose**, which required more initial setup effort, to reduce infrastructure costs by over 90% compared to the expensive **AWS Managed Airflow (MWAA)**. This was also driven by MWAA's version constraints which did not support my preferred **TaskFlow API**.
 - * **Trade-off 2 (Executor):** Chose the simplest and most stable **Sequential Executor** to minimize infrastructure complexity, as the project's core need was sequential stability over the parallel processing offered by the more complex **Celery Executor**.
 - * **Specific Implementation (How):** Deployed Airflow on EC2 using **Docker Compose**. Authored a scraping DAG that calls the scraper management server's API via a **PythonOperator**, and a data purification DAG that triggers an EMR job using **boto3**.
 - **Result & Impact:**
 - * **Maximized Operational Efficiency:** **Fully automated** the daily manual data tasks, reducing the time spent on related operations by over 95%.
 - * **Improved Stability & Observability:** Enabled central monitoring of all pipeline statuses and logs via the Airflow UI, significantly reducing the time to identify and respond to failures.
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Embedding-based Image/Text Deduplication System Development and Optimization

- **Core Problem & Goal:** The initial deduplication system had significant **unnecessary overhead**, as it required downloading embeddings from the DB, converting data types, and creating an index externally. The goal was to **optimize the pipeline** and improve data quality by handling all logic within the database.
- **My Role:** Analyzed the pipeline overhead of the initial system developed by an ML engineer and **led the design and implementation of a new, optimized architecture based on PGVector**.
- **Problem-Solving Process:**
 - * **Technical Consideration (Problem with old way):** The initial **'Faiss-CPU'**-based system was inefficient as it required pulling data from PostgreSQL and **converting incompatible data types** before creating an index locally.
 - * **Specific Implementation (How):** To solve this overhead, I changed the process to handle all logic inside the DB. I performed a data migration from the existing **'Numpy' array** to the **'pgvector::vector' type** and rebuilt the system by creating **'PGVector's 'IVFFlat' and 'HNSW' indexes**.
- **Result & Impact:**
 - * **Improved Operational Efficiency & Cost Savings:** By completely eliminating the unnecessary data transfer and conversion steps, I **reduced the overall pipeline overhead by over 50%** and simplified the architecture.
 - * **Improved Data Quality:** The optimized system dramatically improved data consistency and reliability, playing a key role in building a high-quality Price Database.

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Data Warehouse Redesign for Cost Optimization and Data Reliability

- **Core Problem & Goal:** The existing single RDS-centric architecture suffered from severe **data duplication and unnecessary cost issues**. The goal was to improve data reliability and reduce operational costs by separating storage layers for each data processing stage and introducing an **architecture based on a Single Source of Truth (SSoT)**.
- **My Role:** Analyzed the problems in the data pipeline and **led the redesign of the data warehouse structure** to improve cost-efficiency and data reliability.
- **Problem-Solving Process:**
 - * **Trade-off Analysis (RDS vs. S3 Data Lake):** While the existing RDS structure was simple to develop initially, its long-term cost and scalability issues were clear. I made the decision to **introduce the low-cost S3 as a data lake**, despite the initial migration costs, to secure long-term storage cost savings and management flexibility.
 - * **Specific Implementation (How):** Visually designed DataSource, Data Warehouse, and Data Mart layers using the DBDiagram tool. **Designed approximately 50 table schemas, including an SSoT table** to centrally manage duplicate data.
- **Result & Impact:**
 - * **Established Foundation for Cost Savings:** By introducing an S3 data lake, I established a foundation that could dramatically **reduce anticipated storage costs** compared to the previous architecture.
 - * **Improved Data Reliability:** Established a structure that can systematically manage data duplication and improve consistency and reliability through the **SSoT table design**.

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MLOps Engineer, MLOps at Eazel Inc.

Oct 2021 - Jun 2022

Jung-gu, Seoul

MLOps-based Instagram Network Analysis and Investment Insight PoC

MLOps Engineer

- **Core Problem & Goal:** To objectify/quantify artist market influence, which previously relied on experts' subjective judgments, using Instagram network data. The final goal was to prove growth potential with data and **succeed in securing Series A funding**.
- **My Role:** As an MLOps Engineer, I led the construction of a **reproducible End-to-End pipeline** for the PoC, from data collection, **data/model versioning using DVC**, processing inputs for various models, to K-Means/PCA analysis and result visualization.
- **Problem-Solving Process:**
 - * **Technical Consideration (Why):** Lacking deep knowledge of the machine learning models themselves at the time, I chose to collaborate with an ML Research Engineer, allowing us to focus on our respective areas of expertise. My colleague handled various graph embedding models like **Node2Vec** and **GAT**, while I focused on the data pipeline and subsequent analysis/visualization.
 - * **Specific Implementation (How):** Introduced **DVC (Data Version Control)**, which integrates with Git, to track and version the model's input data and artifacts by linking them to Git commit hashes, thereby **ensuring the reproducibility** of experiments. Afterwards, I took the extracted embedding vectors as input and analyzed artist groups using **K-Means clustering** and **PCA**. Finally, I implemented an interactive network visualization dashboard with these analysis results using **Next.js** and **Cytoscape.js**.
- **Result & Impact:**
 - * **Business Impact:** Proved artist growth potential through the analytics from the developed visualization dashboard, making a **decisive contribution to successfully securing Series A funding**.
 - * **Personal & Technical Growth:** Through this collaborative experience, I gained a deep practical understanding of building machine learning pipelines, and it became the decisive catalyst for my career transition to a Data Engineer.

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Data Scraper Development and Enhancement for Bypassing Instagram Bot Detection

- **Core Problem & Goal:** To solve the problem of being unable to continuously collect data due to Instagram's strong **Bot Detection system**. The goal was to develop a robust scraper that could stably bypass bot detection and collect large-scale social relationship data **without interruption**.
- **My Role:** Responsible for researching, prototyping, and implementing the final solution, exploring various technical approaches (**Selenium**, **requests**, **Instagrapi**, etc.) to solve the bot detection problem.

- **Problem-Solving Process:**
 - * **Trade-off Analysis (Server vs. Mobile Environment):** Confirmed that using a Python-based scraper in a server environment was ultimately blocked by bot detection. Based on the hypothesis that the 'server environment' itself was a major factor in detection, I chose a fundamental solution: to **personally migrate the Python library to Swift**, despite the high initial cost, to emulate requests from a mobile environment.
 - * **Specific Implementation (How):** **Personally ported the core logic of 'Instagrapi' from Python to Swift**, developing a data scraper that runs in an iOS mobile environment. During development, I fixed bugs found in the library and **contributed directly to the open-source project**.
 - **Result & Impact:**
 - * **Core Problem Solved:** Succeeded in **stably collecting large-scale social relationship data**, which was impossible with previous methods, providing the foundation for the 'MLOps Analysis PoC' project.
 - * **Demonstrated Problem-Solving Ability:** Solved a persistent external system limitation through the creative approach of porting a framework to a different language.
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Initial Data Pipeline Construction using PostgreSQL Schema Separation

- **Core Problem & Goal:** Needed to efficiently separate and manage collected raw data and purified data. In a situation where building a separate warehouse was difficult, the goal was to build a practical pipeline that could **preserve data integrity and reliably serve data within a single DB**.
 - **My Role:** Developed the data purification logic and **personally designed and implemented an architecture that separates data layers by separating schemas within PostgreSQL**.
 - **Problem-Solving Process:**
 - * **Trade-off Analysis (DB Separation vs. Schema Separation):** Physically separating the database had the disadvantage of doubling costs and management points. Instead, I chose the **PostgreSQL schema separation method**, which allows for logical separation with no additional cost, securing practicality and cost-efficiency.
 - * **Specific Implementation (How):** Created `raw_data` and `purified_data` schemas within **PostgreSQL**. Built a pipeline that performs data purification logic using **Python's psycopg2 library** and stores the data in tables within their respective schemas.
 - **Result & Impact:**
 - * **Ensured Data Integrity:** Established an environment where raw data is preserved without changes and analysts can safely use only purified data, reducing data management complexity.
 - * **Paved the Way for Growth:** This experience provided a foundational understanding of the need for data warehouses, paving the way for designing more sophisticated data architectures later on.
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iOS Developer, iOS Team

Jan 2019 - Oct 2021
Gangnam, Seoul

iOS Application Development & Deployment

iOS Developer

- **Core Problem & Goal:** To develop a native iOS application for displaying art content, featuring VR capabilities, asset downloading, and AR functionalities.
- **My Role:** Led the development of the app's backend layer, including the API Client. Implemented detail views and view controllers for artworks, artists, and exhibitions. Developed key features including a Google Cardboard VR view via a Unity Bridge, a download service for VR assets, and an AR-based "Artwork Hanger" feature using ARKit.
- **Problem-Solving Process:**
 - * **Technical Consideration (Why):** Faced with upcoming complex features like VR asset downloading and Unity integration, I anticipated that the existing MVC structure would lead to increased complexity and reduced maintainability. The goal was to adopt a reactive architecture for clearer and more predictable state management.
 - * **Trade-off Analysis (Trade-off):** I first considered **implementing the MVVM pattern manually**. However, I judged that this would require significant initial development resources for defining protocols and data binding logic. As an alternative, I reviewed the reactive programming framework, **ReactorKit**. I was impressed that ReactorKit enforces a clear, 3-step unidirectional data flow — **User Input (Action) -> Business Logic Processing (Mutate) -> Applying State Change (Reduce)** — which fundamentally prevents the complexity of manual implementation and makes the code flow very easy to understand. I concluded that despite the initial learning curve, adopting ReactorKit was more advantageous for long-term maintainability and development efficiency.

- * **Specific Implementation (How):** Re-architected the app based on ReactorKit. Unified the data flow by defining all user inputs as 'Actions' and state changes as 'State', and integrated the download service using an AWS S3 client into the Reactor logic.
- **Result & Impact:**
 - * **Improved Development Efficiency:** After getting used to the framework, **the process of defining logic according to the Action-Mutate-Reduce structure became very easy and clear.** This led to improved development speed for new features and maintained code consistency.
 - * **Enhanced Stability & Maintainability:** The clear structure reduced the occurrence of state management-related bugs, and shortened the time it took for new team members to understand the codebase.